



TEST CODE **002372**

FORM TP 23177

MAY/JUNE 2003

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

ADVANCED PROFICIENCY EXAMINATION

BIOLOGY

UNIT 1 – PAPER 02

2 hours

In addition to the 2 hours, candidates are allowed a reading time of 15 minutes. Candidates may write in their answer booklets during this 15-minute period.

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of NINE questions.
2. Section A consists of THREE questions. Candidates must attempt ALL questions in this section and should spend no more than 30 minutes on this section. Answers to this section MUST be written in this answer booklet.
3. Section B consists of SIX questions. Candidates must attempt THREE questions in this section, ONE question from EACH module. Answers to this section MUST be written in the answer booklet provided.
4. The use of silent non-programmable calculators is allowed.

SECTION A

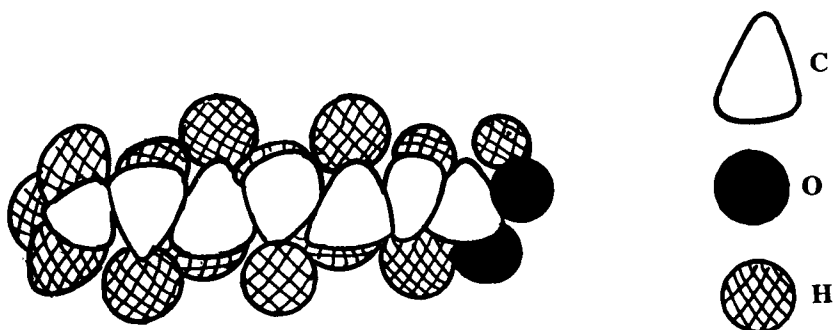
You must attempt ALL THREE questions in this section. You should NOT spend more than 30 minutes on this section.

1. (a) You are writing instructions in a laboratory manual for a practical in which students have to perform a lipid test on an unknown solution. Complete the format below with the appropriate information in each square.

Test for Lipid	Reason for or basis of steps in the test
Steps taken: 1.	
2.	
3.	
4.	
5.	

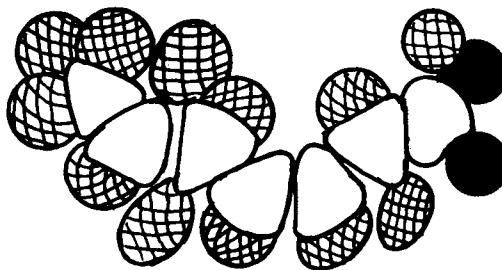
[4 marks]

- (b) The diagrams below show models of two fatty acids. Draw the structural formula for each fatty acid.



Fatty acid No. 1.

- (i) Structural formula for fatty acid No. 1.



Fatty acid No. 2.

- (ii) Structural formula for fatty acid No. 2.

[2 marks]

[2 marks]

GO ON TO THE NEXT PAGE

- (c) State TWO advantages which animals derive from using fat as a storage material other than energy content, in preference to starch.

[2 marks]

Total 10 marks

NOTHING HAS BEEN OMITTED

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2. (a) Figure 1 is a photograph of the embryo sac of a flowering plant, showing nuclear division.

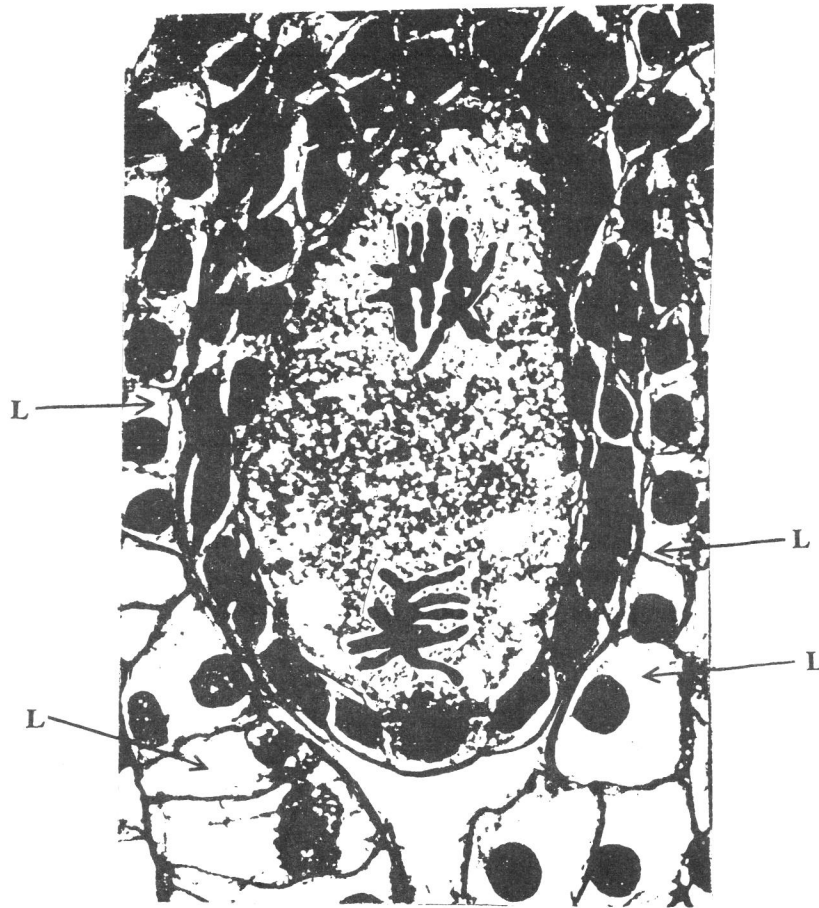


Figure 1

*From An Atlas of Plant Structure Vol 1 by Brian Bracegirdle and Patricia H Miles.
Reprinted by permission of Harcourt Education Publishers.*

- (i) Make an accurate drawing of the **upper group** of chromosomes in Figure 1 above, which are opposite to the micropylar end. Your drawing of the group of chromosomes should be **THREE** times larger than the group in the photograph.

[3 marks]

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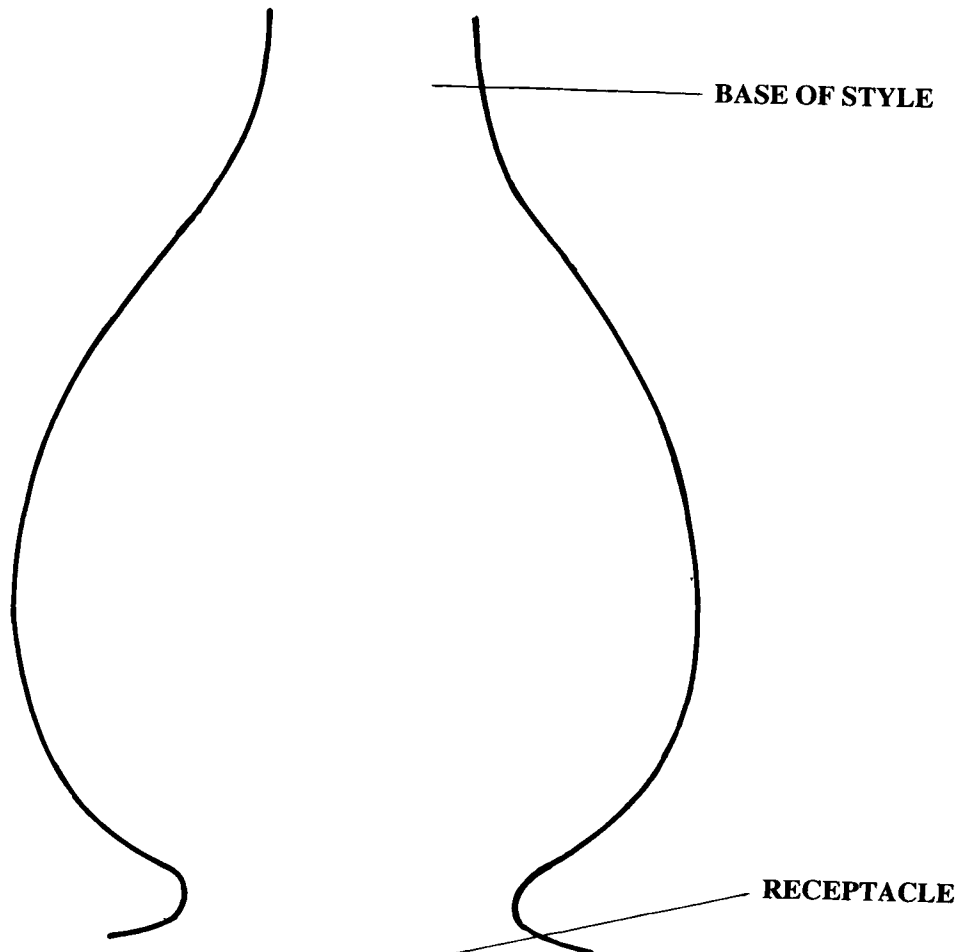
- (ii) State the TYPE of division and the STAGE shown in Figure 1 on page 6 .

[1 mark]

- (b) What happens to the layer labelled L in Figure 1 on page 6, after fertilization?

[1 mark]

- (c) Complete the drawing below to show the contents of the ovary, and indicate by means of a rectangle drawn to scale, in ink, where the photograph of Figure 1 fits into your drawing. Label the rectangle "**Import from Figure 1**".



[3 marks]

- (d) During the development of the embryo sac the original nucleus divides three times to form 8 nuclei. Each of these 8 nuclei has 2 nucleoli, as shown in Figure 2 (a). Some time after pollination, a single nucleus with 6 nucleoli, similar to that shown in Figure 2 (b) can be seen.

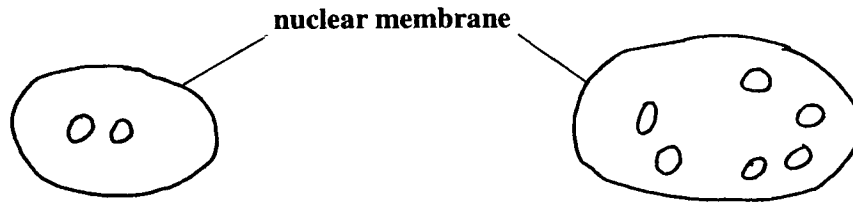


Figure 2 (a)

**A single nucleus in the embryo
sac with 2 nucleoli**

Figure 2 (b)

**A single nucleus in the embryo
sac with 6 nucleoli**

Explain how the 6 nucleoli arise in this nucleus.

[2 marks]

Total 10 marks

NOTHING HAS BEEN OMITTED

3. Figure 3 below is a graph grid that is to be used in answering questions (a) (i) and (ii).

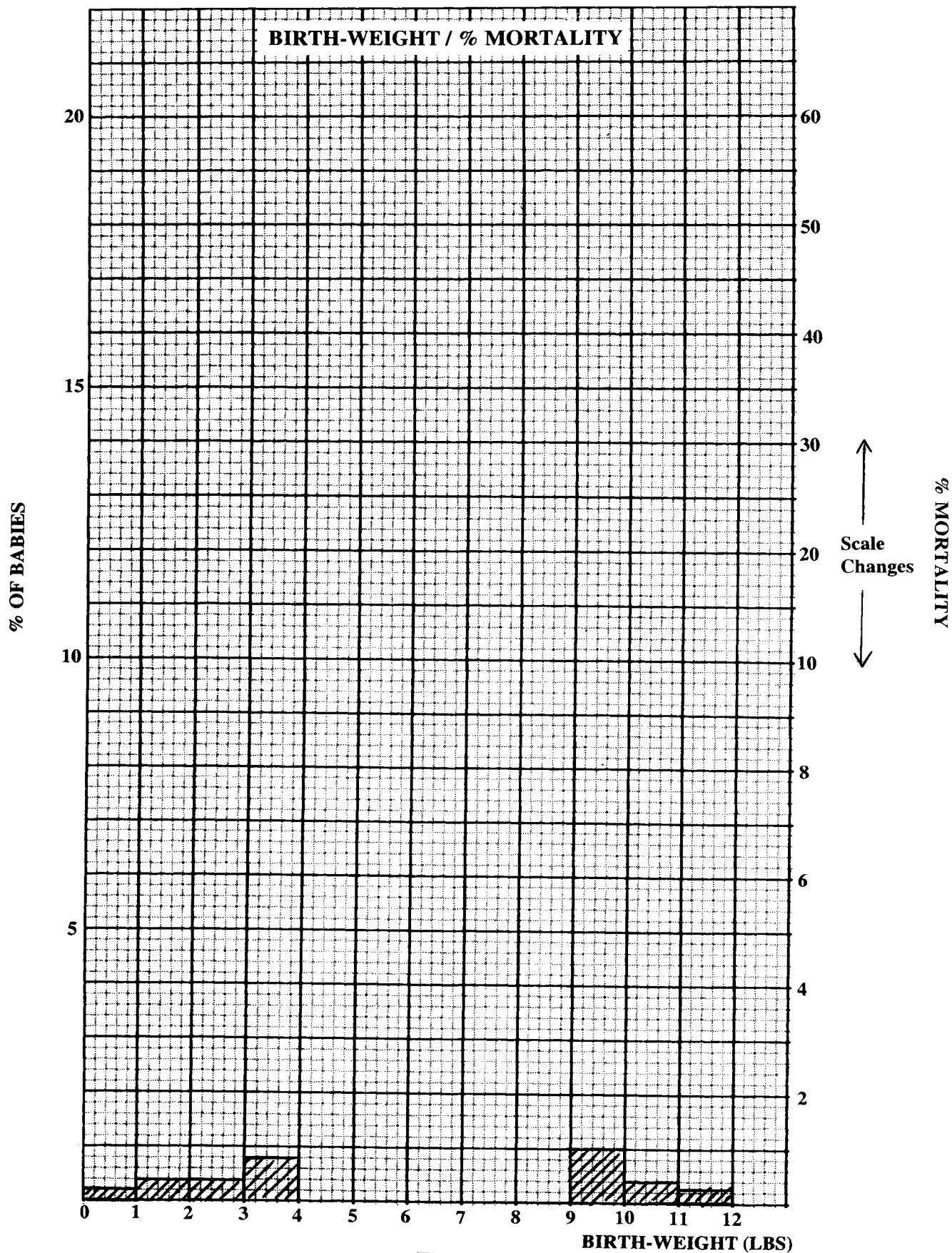


Figure 3

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Table 1 below, contains data on the typical birth-weight of babies over a period of 10 years. The mortality of babies, in relation to birth-weight is shown in Table 2.

TABLE 1. TYPICAL BIRTH-WEIGHT (AS A PERCENTAGE) OVER 10 YEARS

Birth-weight of babies (lbs)	% of babies
0 – 1	} pre-drawn on histogram
1 – 2	
2 – 3	
3 – 4	
4 – 5	2
5 – 6	9
6 – 7	18
7 – 8	14
8 – 9	5
9 – 10	} pre-drawn on histogram
10 – 11	
11 – 12	

TABLE 2. PERCENTAGE MORTALITY OF BABIES IN RELATION TO BIRTH-WEIGHT

Birth-weight of babies (lbs)	% mortality
3	60
4	30
5	7
6	2.5
7	1.5
8	2
9	3
10	6
11	15

Use the data in Tables 1 and 2 above to answer questions (a) (i) and (ii)

- (a) (i) On Figure 3, complete the **histogram** of the typical birth-weight of babies over a 10-year period. (**Note** that portions of this histogram have been pre-drawn.)
- (ii) Also on Figure 3, construct a **graph** of the percentage mortality of babies in relation to birth-weight. (Use a best-fit graph line.) [5 marks]

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- (b) With reference to the graphs drawn on page 10, comment on the effect of birth-weight on mortality.

[3 marks]

- (c) What effect would the observations you made in (b) have on the gene pool controlling the birth-weight of the foetus?

[1 mark]

- (d) What is this type of selection called?

[1 mark]

Total 10 marks

SECTION B

You must answer THREE questions in this section. Answer ONE question EACH from Modules 1, 2 and 3. You MUST write your answers in the answer booklet provided.

MODULE 1

Answer EITHER Question 4 OR Question 5.

- 4.** (a) State briefly what is meant by the primary, secondary, tertiary and quaternary structure of proteins. Exemplify EACH answer by reference to the protein molecule, collagen. [10 marks]
- (b) The structure of collagen confers special properties which reflect its distribution in the body. Comment on this statement. [6 marks]
- (c) A species of toad may produce eggs which are unable to form nucleoli. Discuss the outcome of this deficiency on the function of
- (i) cellular ribosomes
 - (ii) mitochondrial ribosomes. [4 marks]

Total 20 marks

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5. (a) Describe the structure of glucose, starch and cellulose and explain how these compounds function in living organisms. **[10 marks]**
- (b) One goldfish is placed in each of two goldfish bowls filled with water. One goldfish bowl is placed in a freezer at 0°C and the other is placed in blazing sunshine. After one hour, the gold fish bowl is removed from the freezer and the water is seen to have a thin layer of ice on the top. The bowl in the sunshine has lost a small quantity of water by evaporation.
- Discuss how the properties of water have enabled EACH goldfish to survive in the contrasting conditions. **[5 marks]**
- (c) The ruby-throated humming bird stores fat before migrating across the Gulf of Mexico. Fats, for example triglycerides, contain twice as much energy, per gram, as carbohydrates, for example sucrose. Fat is, therefore, an efficient energy storage material for animals.
- Using a comparison of the way fats and carbohydrates interact with water, suggest why fat is the BETTER energy storage material for the humming bird. **[5 marks]**

Total 20 marks

MODULE 2

Answer EITHER Question 6 OR Question 7.

6. (a) Using **named** flowering plants, describe TWO different methods used to promote cross pollination. **[6 marks]**
- (b) Describe the features of the human egg which enable it to carry out its function. **[4 marks]**
- (c) The uterus provides a safe place where the foetus can develop. Compare the contributions the amnion and the placenta make to the safety and development of the foetus. **[10 marks]**
- Total 20 marks**
7. (a) Describe the roles of DNA, mRNA, rRNA and tRNA in the processes of transcription and translation. **[10 marks]**
- (b) In Werner' syndrome, individuals age prematurely and often die of age related conditions before the age of 50. These individuals have a mutation in the gene that codes for protein enzymes that are responsible for DNA replication. Suggest why this mutation has such severe effects on the individual. **[4 marks]**
- (c) An intercalating agent is a dye that can insert itself between base pairs of DNA. Discuss the effect this compound would have on DNA and its replication and predict how RNA and its formation might be affected by this agent. **[6 marks]**

Total 20 marks

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MODULE 3

Answer EITHER Question 8 OR Question 9.

8. (a) Describe the biological-species concept and state the limitations of this concept. Explain how new species are formed. **[10 marks]**
- (b) Opposing environmental pressures may give rise to balanced polymorphism, in which two or more alleles are maintained in a population because each is favoured by a separate environmental force. Suggest how environmental factors act to preserve the alleles for both normal haemoglobin and sickle cell haemoglobin in native Africans living in Africa. **[5 marks]**
- (c) Discuss the advantages of sexual reproduction over asexual reproduction. **[5 marks]**

Total 20 marks

9. (a) Give an account of FIVE methods that may be used to conserve useful or valuable species. **[10 marks]**
- (b) Flower colour in Rosepea may be red (dominant), or pink, (recessive). The petals usually have a plain border (dominant), but in 2001, a mutant pink flowered plant was found which had frilled petals, a **recessive feature**.

You are a plant breeder with access to red plain homozygous, and pink frilled plants. Design a series of genetic crosses to demonstrate theoretically, how you could obtain plants which produce only red frilled flowers. **[10 marks]**

Total 20 marks

END OF TEST

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